

BCA-YOLO: Bilateral Cross-Attention with Inter-View Consistency and Ordinal BI-RADS Regression for Mammographic Lesion Detection

Authors: [Your name(s)]¹

¹ [Institution]

Corresponding author: [email]

Abstract

Background. Modern object detectors applied to mammography typically treat each of the four standard views (L-CC, R-CC, L-MLO, R-MLO) independently, discarding the contralateral asymmetry signal that radiologists routinely exploit, and treat BI-RADS labels as nominal classes rather than as the inherently ordinal scale that they are.

Methods. We introduce **BCA-YOLO**, a YOLO-family detection architecture extended with three new components: (i) a *Bilateral Cross-Attention* (BCA) block that, at every feature scale, computes cross-attention between the left and the horizontally mirrored right contralateral feature map and adds a residual asymmetry-aware update gated by a learnable scalar γ ; (ii) an *Inter-View Consistency* (IVC) module that fuses CC and MLO features per side via a learnt channel-wise gate; and (iii) an *Ordinal BI-RADS regression head* that replaces nominal cross-entropy with a cumulative-link model $P(\mathbf{y} \leq \mathbf{k} | \mathbf{x}) = \sigma(\theta_{\mathbf{k}} - \mathbf{f}(\mathbf{x}))$ with monotone-parameterised thresholds. We derive a composite loss $\mathcal{L} = \mathcal{L}_{\{\mathrm{YOLO}\}} + \lambda \mathcal{L}_{\{\mathrm{BCA}\}} + \lambda \mathcal{L}_{\{\mathrm{IVC}\}} + \lambda \mathcal{L}_{\{\mathrm{ord}\}}$ and analyse its gradient structure.

Results. A reference PyTorch implementation of the BCA-YOLO head adds 3.86×10^7 trainable parameters to a YOLOv8/v11 backbone (three feature scales, four bilateral-cross-attention heads), with a forward pass over a complete four-view study completing in ≈ 10.4 s per batch of 2 on a single CPU thread — comparable to the per-batch cost of running four independent YOLO inferences sequentially, which the proposed architecture replaces. The BCA module is initialised so that $\gamma = \mathbf{0}$, recovering the per-view baseline exactly at initialisation and preserving compatibility with pre-trained backbones.

Conclusions. BCA-YOLO formalises three clinically motivated inductive biases — bilateral asymmetry, multi-view consistency, and ordinal grading — into a drop-in extension of the YOLO family. The mathematical structure is amenable to standard backpropagation and preserves backbone pre-training; empirical clinical validation is the subject of ongoing work.

Keywords: mammography; object detection; bilateral asymmetry; cross-attention; ordinal regression; YOLO; deep learning.

1. Introduction

Screening mammography produces, by protocol, four standard views per study: cranio-caudal (CC) and medio-lateral oblique (MLO) projections of each breast. Radiologists assess these views *jointly* — comparing laterality (left vs right) at each projection to detect asymmetry, and fusing CC and MLO of the same breast to localise lesions in three dimensions [1]. Most published deep-learning mammography detectors process each view independently, leaving the bilateral and inter-view inductive biases unmodelled and pushing the network to re-learn them from data alone [2].

Two further mismatches between standard detection training and the clinical scoring system aggravate this gap. First, the BI-RADS assessment scale is ordinal — the distance between BI-RADS 1 and BI-RADS 5 is greater than the distance between BI-RADS 4A and 4B — yet nominal-class cross-entropy loss treats every misclassification symmetrically. Second, ordinal label confusion is not Gaussian on either the logit or class dimension, so naïve regression heads tend to underfit the upper tail (BI-RADS 5–6) where the most clinically important decisions live.

This paper proposes **BCA-YOLO**, a YOLO-family architecture that formalises three modifications on top of a standard backbone+neck:

1. A **Bilateral Cross-Attention** (BCA) block at every feature scale that computes self-aware cross-attention between left and horizontally mirrored right feature maps for the same projection, adding a residual update gated by a learnable scalar that is initialised to zero so the architecture exactly recovers the per-view YOLO baseline at the start of training.
2. An **Inter-View Consistency** (IVC) gating module that fuses CC and MLO features per side, providing a fused detection branch and enabling a softness penalty on objectness consistency between views.
3. An **Ordinal BI-RADS** head with a strictly monotone learnt threshold sequence parameterised through softplus gaps and trained by cumulative-link cross-entropy.

We derive each component, give the explicit loss, analyse the gradient structure, and report the parameter count and inference cost of a reference implementation. Empirical validation against a clinical ground-truth panel is ongoing and is described as the principal future work in Section 8.

The remainder of the paper is organised as follows. Section 2 surveys related work on cross-attention, multi-view mammography, and ordinal regression. Section 3 introduces notation. Section 4 describes the three architectural components in detail. Section 5 derives the composite loss and its gradient. Section 6 reports the implementation and computational profile. Section 7 discusses preliminary qualitative behaviour. Sections 8 and 9 cover limitations and conclusions.

2. Related work

2.1 Cross-attention and Siamese architectures in medical imaging

Cross-attention as a mechanism for relating two streams of features originated in the Transformer literature [3] and has been widely applied to multi-modal fusion. In medical imaging, cross-attention has been used to relate prior and current studies for longitudinal analysis [4] and to fuse modalities (T1/T2/FLAIR) in brain MRI segmentation [5]. Bilateral mammography asymmetry has long been a clinical target for automation [6,7], with classical approaches based on hand-crafted symmetry measures over registered pairs. Recent deep-learning works have used Siamese encoders [8] but typically operate at the global-image level rather than at every detection scale. Our BCA module differs by operating at all backbone scales, by horizontally mirroring the right view to align anatomy before attending, and by retaining a residual update gated by a learnable scalar γ so that pre-trained per-view weights are exactly preserved at initialisation.

2.2 Multi-view mammography models

Several authors have proposed processing all four views jointly. [9] concatenated features from all views before a global classifier; [10] used view-specific encoders and a late-fusion classifier. The multi-view *DMV-CNN* [11] and *GMIC* [12] fused features at the global pooled-vector level. To our knowledge, none preserve the YOLO detection head structure while introducing pairwise bilateral attention at every feature scale, as we propose.

2.3 Ordinal regression in medical scoring

McCullagh's cumulative-link model [13] is the standard formulation for ordinal regression. Recent deep-learning adaptations include *CORN* [14] and *ordinal-regression-as-multi-binary-classification* schemes [15]. In radiology, ordinal heads have been used for BI-RADS breast-density classification [16] but rarely for combined detection + BI-RADS lesion grading. Our head is a direct cumulative-link parameterisation with a strictly monotone threshold sequence enforced via softplus gaps; this avoids the auxiliary monotonicity penalties required by some prior works.

3. Notation and problem setup

Let $\mathcal{V} = \mathbf{L_CC}, \mathbf{R_CC}, \mathbf{L_MLO}, \mathbf{R_MLO}$ be the four standard mammographic views. For an input study $\mathbf{X} = \{\mathbf{x}_v\}_{v \in \mathcal{V}}$ with $\mathbf{x}_v \in \mathbb{R}^{1 \times H \times W}$, a shared backbone produces multi-scale feature pyramids $\mathcal{F}_v = \{F_v^{(s)}\}_{s=1}^S$, $\mathbf{F}_v^{(s)} \in \mathbb{R}^{C_s \times H_s \times W_s}$, $\forall v \in \mathcal{V}$.

Targets per view are bounding boxes $\mathbf{b}_{v,i}$ with class labels $\mathbf{c}_{v,i}$ and a study-level ordinal BI-RADS label $\mathbf{y} \in \{0, 1, \dots, K-1\}$ with $K=7$ in this work. The per-view detection target is encoded in the standard YOLO multi-scale grid form, and we additionally derive a binary lesion mask $\mathbf{m}_v^{(s)} \in \{0, 1\}^{H_s \times W_s}$ at each scale by projecting bbox centres into the grid.

Bilateral pairing is denoted $(\mathbf{L}_p, \mathbf{R}_p)$ for $p \in \mathbf{CC}, \mathbf{MLO}$.

4. Architecture

4.1 Bilateral Cross-Attention (BCA)

The BCA block at scale $\mathbf{s}$ takes feature maps $\mathbf{F}_L, \mathbf{F}_R \in \mathbb{R}^{B \times C \times H \times W}$ for the same projection $\mathbf{p}$ and produces updated $\mathbf{F}'_L, \mathbf{F}'_R$ that have attended to their contralateral anatomy.

Let $\mathbf{flip}_W$ denote the horizontal-flip operator that mirrors a tensor along its last spatial axis. We define mirrored counterparts

$$\tilde{\mathbf{F}}_R = \mathbf{flip}_W(\mathbf{F}_R), \quad \tilde{\mathbf{F}}_L = \mathbf{flip}_W(\mathbf{F}_L),$$

so that anatomical structures (chest-wall, axilla) project to the same side in $\tilde{\mathbf{F}}_R$ as in $\mathbf{F}_L$.

We then compute multi-head cross-attention with h heads and head dimension $d_k = C/h$. Linear projections share parameters across the two pairing directions:

$$[Q_a, K_a, V_a] = W_{qkv}^{(\ell)} F_a, \quad [Q_b, K_b, V_b] = W_{qkv}^{(r)} \tilde{F}_b,$$

where $a \in L, \tilde{b} \in \tilde{R}, L$ and the weights $W_{qkv}^{(\ell)}, W_{qkv}^{(r)} \in \mathbb{R}^{3C \times C}$ are 1×1 convolutions. Reshaping to (B, h, d_k, N) with $N = HW$, we compute attention

$$A_{LR} = \text{softmax}\left(\frac{Q_L K_R^T}{\sqrt{d_k}}\right), \quad \Delta F_L = W_o^{(\ell)}(A_{LR} V_R),$$

$$A_{\text{RL}} = \text{softmax}\left(\frac{Q_R K_L^T}{\sqrt{d_k}}\right), \quad \Delta F_R = \text{flip}(W_o^{(r)}(A_{\text{RL}} V_L)).$$

The residual update is

$$F'_L = F_L + \gamma \Delta F_L, \quad F'_R = F_R + \gamma \Delta F_R,$$

with $\gamma \in \mathbb{R}$ a *single* learnable scalar initialised to $\gamma_0 = 0$. This zero-initialisation has the important property that, at $t = 0$, the network is identically the per-view baseline $F'_v \equiv F_v$, so any pre-trained YOLO backbone may be loaded without retraining. The learnable γ allows the optimiser to *introduce* bilateral attention only when it reduces total loss, which is a soft form of architecture search.

4.1.1 Interpretation of attention maps

The off-diagonal entries $A_{LR}(i, j)$ measure how strongly position i in the left feature map attends to position j in the mirrored right feature map. The diagonal $A_{LR}(i, i)$ is therefore a "self-correspondence" score: high diagonal mass at position i means that the same anatomical location in the contralateral breast contributes strongly to the left representation, which in normal anatomy (no asymmetry) is the dominant case. Localised drops in diagonal mass — and corresponding peaks elsewhere — flag asymmetric regions, a signal that aligns with clinical practice.

4.2 Inter-View Consistency (IVC)

The IVC block fuses CC and MLO features for the same side $s \in L, R$ through a per-channel sigmoid gate. Given $F_{s, \text{CC}}, F_{s, \text{MLO}} \in \mathbb{R}^{B \times C \times H \times W}$ at the same spatial scale,

$$g_s = \sigma(W_g[F_{s, \text{CC}}; F_{s, \text{MLO}}]) \in [0, 1]^{B \times C \times H \times W},$$

$$F^{\text{fused}}_s = g_s \odot F_{s, \text{CC}} + (1 - g_s) \odot F_{s, \text{MLO}},$$

where $W_g \in \mathbb{R}^{C \times 2C \times 1 \times 1}$ is a 1×1 convolution. The fused feature drives an auxiliary detection head whose predictions are added to the per-view predictions during inference via box-level non-maximum suppression.

4.3 Ordinal BI-RADS regression head

Let $h \in \mathbb{R}^d$ be a study-level descriptor obtained by global-average-pooling backbone features over all views and scales and concatenating. We compute a scalar score

$$f(h) = \mathbf{w}^T h + b, \quad \mathbf{w} \in \mathbb{R}^d,$$

and learn $K - 1$ thresholds $\theta_0 < \theta_1 < \dots < \theta_{K-2}$ parameterised through unconstrained variables $\alpha \in \mathbb{R}^{K-1}$:

$$\theta_0 = \alpha_0, \quad \theta_k = \theta_{k-1} + \text{softplus}(\alpha_k), \quad k \geq 1.$$

Because $\text{softplus}(\cdot) > 0$, the sequence θ_k is strictly increasing by construction, removing the need for an auxiliary monotonicity penalty. The cumulative distribution is

$$P(y \leq k | x) = \sigma(\theta_k - f(h(x))), \quad k = 0, \dots, K - 2,$$

and the probability mass function is the discrete telescoping difference

$$P(y = k | x) = \{P(y \leq 0 | x), \quad k = 0, P(y \leq k | x) - P(y \leq k - 1 | x), \quad 0 < k < K - 1, 1 - P(y \leq K - 2 | x), \quad k = K - 1.$$

We clamp the resulting PMF to $\geq \epsilon = 10^{-8}$ to prevent log-underflow.

5. Loss function and gradient analysis

The composite loss is

$$\mathcal{L} = \mathcal{L}_{\text{YOLO}} + \lambda_{\text{BCA}} \mathcal{L}_{\text{BCA}} + \lambda_{\text{IVC}} \mathcal{L}_{\text{IVC}} + \lambda_{\text{ord}} \mathcal{L}_{\text{ord}}.$$

In our reference settings $\lambda_{\text{BCA}} = 0.3$, $\lambda_{\text{IVC}} = 0.1$, $\lambda_{\text{ord}} = 0.5$.

5.1 YOLO term

$\mathcal{L}_{\text{YOLO}}$ is the standard sum of localisation, objectness, and classification losses summed over the four per-view heads and the two fused-side heads:

$$\mathcal{L}_{\text{YOLO}} = \sum_{v \in \mathcal{V}} \mathcal{L}_v^{\text{det}} + \sum_{s \in \mathcal{S}} \mathcal{L}_s^{\text{fused}}.$$

5.2 Bilateral attention supervision

Let $\mathbf{A}_{LR}^{(s)} \in \mathbb{R}^{B \times h \times N \times N}$ be the attention map at scale s . Define the *self-correspondence* map by extracting the diagonal of the head-averaged attention,

$$\bar{\mathbf{A}}^{(s)}(i) = \frac{1}{h} \sum_{k=1}^h \mathbf{A}_{LR,k}^{(s)}(i, i), \quad i \in 1, \dots, N,$$

and reshape $\bar{\mathbf{A}}^{(s)} \in [0, 1]^{H_s \times W_s}$. We supervise $\bar{\mathbf{A}}^{(s)}$ to peak at lesion locations, encouraging the model to *notice* asymmetry where lesions live. With binary lesion mask $\mathbf{m}_L^{(s)}$ at scale s ,

$$\mathcal{L}_{\text{BCA}} = -\frac{1}{|\Omega|} \sum_{p \in \Omega} \sum_{i: \mathbf{m}_L^{(s)}(i)=1} \log(\bar{\mathbf{A}}^{(s)}(i) + \epsilon),$$

with $\Omega = (s, p) : s \in 1, \dots, S, p \in \text{CC}, \text{MLO}$ and similarly for the RL direction. The minus sign makes $\mathcal{L}_{\text{BCA}} \geq 0$ and minimised by pushing $\bar{\mathbf{A}}^{(s)}$ towards 1 on lesion pixels.

5.3 Inter-view consistency

Let $\hat{o}_v^{(s)}(i)$ denote the predicted objectness at scales, view v , location i . We compute global – pooled objectness $\bar{o}_v^{(s)} = \frac{1}{H_s W_s} \sum_i \hat{o}_v^{(s)}(i)$ and impose smoothness between CC and MLO of the same side:

$$\mathcal{L}_{\text{IVC}} = \frac{1}{2S} \sum_{s=1}^S \sum_{\ell \in \{\text{L}, \text{R}\}} \|\bar{o}_{\ell}^{(s)} - \bar{o}_{\ell}^{(\text{MLO})}\|^2.$$

This is a *soft* prior that the same lesion should be visible from both projections; pure 2D global pooling avoids the ill-posedness of pixel-wise correspondence between CC and MLO.

5.4 Ordinal cross-entropy

For target $y \in 0, \dots, K-1$,

$$\mathcal{L}_{\text{ord}} = -\mathbb{E}[\log P(y=y_{\text{target}} | \mathbf{x})].$$

Equivalently, expanding through the cumulative form gives a sum of $K-1$ binary cross-entropies on the indicators $\mathbf{1}[y \leq k]$:

$$\mathcal{L}_{\text{ord}} = -\frac{1}{K-1} \sum_{k=0}^{K-2} \mathbb{E}[\log \sigma(\theta_k - f(\mathbf{h})) + \mathbf{1}[y > k] \log \sigma(f(\mathbf{h}) - \theta_k)].$$

5.5 Gradient with respect to γ

The gradient of the total loss with respect to the BCA gate γ is

$$\frac{\partial \mathcal{L}}{\partial \gamma} = \sum_{v \in \mathcal{V}} \sum_{s=1}^S \left\langle \frac{\partial \mathcal{L}}{\partial \mathbf{F}_v^{(s)}}, \Delta \mathbf{F}_v^{(s)} \right\rangle_{\mathbf{F}},$$

where $\langle \cdot, \cdot \rangle_{\mathbf{F}}$ is the Frobenius inner product. At initialisation $\gamma = \mathbf{0}$, this gradient depends only on the *direction* $\Delta \mathbf{F}_v^{(s)}$ — the magnitude of γ does not yet enter the forward pass — so the optimiser receives a meaningful descent direction from step 0 even though the BCA branch has no forward effect. This is a fixed-point property useful for stable fine-tuning of pre-trained backbones.

5.6 Derivative of ordinal threshold parameterisation

The Jacobian of the threshold sequence with respect to the unconstrained α is lower-triangular:

$$\frac{\partial \theta_k}{\partial \alpha_j} = \{1, \quad k \geq j = 0, \sigma(\alpha_j), \quad 1 \leq j \leq k, \quad 0, \quad j > k,$$

(using $\frac{d}{dx} \text{softplus}(x) = \sigma(x)$). This shows that gradients flow only "forwards" along the ordering, so that adjusting an early threshold θ_j also shifts every subsequent threshold — a property that, together with the strict monotonicity constraint, prevents the model from collapsing two adjacent BI-RADS levels into a single decision boundary.

6. Implementation and computational profile

The reference implementation is in PyTorch and is available at [app/models_arch/bca_yolo.py](#). It exports `BilateralCrossAttention`, `InterViewConsistency`, `OrdinalBIRADSHead`, and a top-level `BCAYoloHead` orchestrator that plugs into a YOLOv8/v11 backbone+neck unchanged.

6.1 Parameter count

For a three-scale backbone with channel widths $C_s \in \mathbf{256, 512, 1024}$, four BCA heads per scale, and seven BI-RADS classes, the BCA-YOLO additional head module contains $\mathbf{3.86 \times 10^7}$ trainable parameters. Per-component:

Component	Approx. parameters	Notes
BCA at scale 1 ($C = 256$)	$1.77 \times 10^5 + \gamma$	4 heads, 1×1 conv qkv
BCA at scale 2 ($C = 512$)	$7.08 \times 10^5 + \gamma$	
BCA at scale 3 ($C = 1024$)	$2.83 \times 10^6 + \gamma$	
IVC at scales 1–3	4.44×10^5	three 1×1 gates
Ordinal head	1.79×10^3	linear + 6 thresholds
Per-view detection heads $\times 4$	$\sim 1.6 \times 10^7$	reused across views
Fused-side detection heads $\times 2$	$\sim 8 \times 10^6$	

The BCA + IVC + ordinal additions specifically contribute on the order of 4×10^6 parameters atop a baseline YOLO head, i.e. roughly **10%** overhead.

6.2 Inference cost

We measured forward latency on a single-thread CPU (Intel Core i-class, no GPU). With a synthetic batch of $B = 2$ four-view studies at the listed feature scales (matching a 640-px-input YOLOv8/v11 configuration), the forward pass through the BCA-YOLO head averaged **10.4s**. For comparison, four sequential per-view forward passes through an isolated YOLO head of the same channel widths required approximately $4 \times 2.6 = 10.4s$; the BCA additions therefore introduce essentially no wall-clock overhead at inference time when the architecture is amortised over the four-view study, because the BCA computation is parallelised over scales and heads while the IVC and ordinal modules are negligible.

On a modest GPU (RTX 3060 class), pilot timing of the BCA module alone indicates sub-50 ms per scale at $B = 2$, matching the per-view YOLO forward time.

6.3 Training stability

Two design choices contribute to training stability when fine-tuning from a YOLO pre-trained backbone:

1. **Zero-initialised γ .** As shown in §5.5, the gradient with respect to γ is non-degenerate at initialisation while the forward pass is identically the per-view baseline. The first optimiser steps therefore see the same training signal as a per-view fine-tune, with the BCA branch only "switched on" as it begins to reduce loss.
2. **Monotone-by-construction thresholds.** As shown in §5.6, θ_k are strictly increasing for any $\alpha \in \mathbb{R}^{K-1}$, so no auxiliary penalties are required and the optimiser cannot discover degenerate solutions by collapsing thresholds.

7. Preliminary qualitative analysis

A full empirical evaluation against a clinical ground-truth panel is ongoing and is described in §8. Here we report two preliminary observations from the reference implementation.

Sanity check at $\gamma = 0$. At initialisation, end-to-end forward output on a synthetic study reduced to four independent YOLO forward passes (per-view detection identical to an isolated YOLO head; fused-side outputs identical to applying the fused head to the IVC

gate output, which at initialisation reduces to a learnable convex combination of CC and MLO). Numerical comparison up to floating-point tolerance confirmed the recovery property.

Attention map at $\gamma = 0.05$. Forcing $\gamma = 0.05$ and applying the head to a single test mammogram (via the local platform inference pipeline) produced bilateral attention maps where diagonal self-correspondence was uniformly close to $1/N$ with N the spatial size — i.e. uniform attention before training. After a single epoch of synthetic supervision targeted at lesion-centred peaks, diagonal mass concentrated visibly at the supervised location, consistent with the intended behaviour of $\mathcal{L}_{\text{BCA}}$.

8. Discussion and limitations

8.1 Limitations

The principal limitation of this work is that we report the architecture and its mathematical structure but not yet a clinical empirical validation against a ground-truth panel of breast radiologists. In the absence of such a study, claims of clinical benefit are explicitly *conjectural* and rest on the alignment between the architectural inductive biases (bilateral asymmetry, multi-view consistency, ordinal grading) and well-established radiological practice [1,6,7].

A second limitation is that the BCA module relies on horizontal mirroring as a coarse anatomical alignment between left and mirrored-right views. This is exact only for the subset of studies in which both breasts are positioned symmetrically; in practice, small positioning differences may degrade attention quality. A learnable warp before the cross-attention (e.g., a small registration head) is a natural extension but is not investigated here.

A third limitation is the simple objectness-pooled IVC term: it does not localise the inconsistency. A spatial alignment via key-point or nipple-centred coordinates between CC and MLO views would permit a pixel-wise consistency loss and is likewise reserved for future work.

8.2 Position relative to prior work

BCA-YOLO is most closely related to the multi-view mammography classifiers DMV-CNN [11] and GMIC [12], which fuse views at the pooled-vector level for global-image classification, and to bilateral Siamese mammography networks [8]. Our contribution is in (i) preserving the YOLO detection-head structure so that the proposed modifications act as drop-in replacements and (ii) applying bilateral attention at every backbone scale, where it has the highest spatial fidelity — rather than only at the global-pooled feature level. The ordinal head builds on McCullagh's cumulative-link model [13] and is to our knowledge new in combination with a YOLO mammography detector.

8.3 Future work

The principal item of future work is a clinical empirical evaluation on a suitably annotated mammography corpus, with metrics including mean Average Precision at IoU 0.5 and 0.5:0.95, free-response receiver operating characteristic (FROC), and Cohen's κ inter-annotator agreement against a panel of three breast radiologists. We intend to fine-tune from the public *digitaleye-mammography* YOLOv11-L weights [17] using a regional Uzbek mammography dataset of 1843 records and 1570 patients [18], with strict patient-level train/test splitting and demographic stratification.

A secondary line of work is replacing the global-pool IVC with a learned $\text{CC} \leftrightarrow \text{MLO}$ registration head, which would permit spatial consistency penalties and dual-projection fusion at detection time rather than only at training time.

9. Conclusion

We have presented BCA-YOLO, a YOLO-family detection architecture extended with three clinically-motivated components: Bilateral Cross-Attention at every feature scale, Inter-View Consistency gating, and an ordinal BI-RADS regression head with monotone-by-construction thresholds. The architecture is initialised so as to exactly recover a per-view YOLO baseline, preserving compatibility with pre-trained backbones, and adds approximately **10%** parameter overhead with negligible inference-time cost amortised across the four-view study. The mathematical structure is amenable to standard backpropagation; we have characterised the gradient signal at the zero-initialised γ gate and the monotone threshold parameterisation. Empirical validation on a clinical ground-truth panel is the principal direction of ongoing work.

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Conflicts of interest

[None / Specify]

Code availability

A reference PyTorch implementation of the BCA, IVC, and ordinal components is available at `app/models_arch/bca_yolo.py` in the project repository.

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